Fiscal Year:	FY 2009	**Task Last Updated:**	FY 04/02/2009
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Neurochemical and Behavioral Effects of Exposure to Heavy Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2007	**End Date:**	12/31/2008
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Sept 2007--Received NCE to 12/31/2008 per S. Krenek/JSC; original end date was 12/31/2007		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Joseph, James (USDA, HNRCA at Tufts University) Shukitt-Hale, Barbara (USDA, HNRCA, Tufts University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Future missions in space (such as a mission to Mars) will involve long-term travel beyond the magnetic field of the earth, subjecting astronauts to radiation hazards posed by solar flares and galactic cosmic rays. Cosmic rays are composed of protons, alpha particles and particles of high energy and charge (HZE particles). Previous research has shown that exposure to HZE particles can affect neural functioning and behavior. These changes may affect the ability of astronauts to successfully meet mission requirements. The objectives of the proposed research program are: (1) to further characterize HZE particle-induced behavioral and neurochemical deficits; (2) to determine the neural mechanisms underlying the changes in performance; and (3) to determine how individual characteristics (age and gender) may influence the responsiveness to HZE particle irradiation. To achieve these objectives, we propose using behavioral and neurochemical measures that have been previously shown to be sensitive to exposure to HZE particles and which may provide a basis for defining the risks of exposure to heavy particles on brain functioning and related behavior.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Effects of Exposure to Different Particles: Based on our previous research with a range of energies of 56Fe particles, we had assumed that the acute effects of exposure to a variety of HZE particles would show the LET-dependence that we had observed following exposure to 56Fe particles. The results of the research reported here does not completely support that assumption. Despite the fact that the LET of 28Si (600 and 1000 MeV/n; LET ˜51 and 44 keV/µm, respectively), 48Ti particles (1100 MeV/n; LET ˜106 keV/µm) and 12C (290 MeV/n; LET ˜13 keV/µm) is significantly less than that of 600 and 1000 MeV/n 56Fe particles (LET ˜189 keV/µm and 149 keV/µm, respectively), the thresholds for the disruption of neural and behavioral functioning following exposure to these particles was significantly less than that following exposure to 56Fe particles. Additional work this year with 16O particles (1000 MeV/n; LET = 14 keV/µm) was consistent with the data presented above. The threshold for disruption of neurocognitive performance was similar to that of 12C. However, this research also indicated that irradiation with different particles and energies did not produce consistent decrements in all measured endpoints. As such, the specific neural and behavioral endpoint may also be a significant factor in determining the effects of exposure to HZE particles. Overall, these results suggest that the relative effectiveness of exposure to different particles may vary as a function of both the particular particle and the specific endpoint under consideration. The data obtained thus far would suggest that it may not be possible to determine the acute effects of exposure to the range of HZE particles to which astronauts may be exposed on exploratory class missions by simply studying the effects of exposure to a single particle or measuring a single endpoint. Effects of Age: The data presented in the present report is consistent with previous research using 56Fe particles which had shown that exposure to a dose of 150 cGy of 56Fe particles did not have acute effects on the ability of the rat to respond appropriately on an ascending fixed-ratio schedule when tested several months following irradiation. However, when tested 12 months following exposure, the irradiated rats fail to increase their rate of responding as the number of responses to obtain reinforcement is increased. The data presented here report similar effects are obtained following exposure to other HZE particles, 28Si, 48Ti and 12C, and with different behavioral endpoints including object recognition memory. Although additional research using lower doses of these particles is needed to establish an absolute threshold in older organisms, the present results are consistent with the hypothesis that there is a continuing performance decrement following exposure to HZE particles as a function of the time following irradiation. Exposing an organism to HZE particles produces long-term changes in the neural systems that underlie the performance of specific behaviors. Because astronauts are likely to be middle aged, the results of these experiments suggest that they may be more susceptible to the effects of exposure to HZE particles on CNS function and on behaviors that depend upon the integrity of the CNS. Radiation-Induced Changes in Neural Structure: The final series of experiments evaluated the effects of exposure to 150 cGy of 56Fe (1000 MeV/n) particles. This dose was selected because it produces significant decreases in spatial learning and memory, a task which is dependent upon the integrity of the hippocampus. Exposing rats to 56Fe particles produced a significant decrease in dendritic branching in Layers II and III of the cortex, but did not affect the number of dentritic spines in the cortex. However, irradiation did cause a significant decrease in the number of M-type (mushroom-type) spines on the dendrites of granule cells in the dentate gyrus of the hippocampus. The results of this experiment indicate that there are changes in the expression of dendrites in the hippocampus as a function of exposure to HZE particles and these changes are related to changes in performance. Summary: In summary, the results indicate that: * The neurobehavioral effects of exposure to different heavy particles resulting from both acute and late degenerative effects of irradiation on CNS function may differ as a function of particle energy, fluence and the specific endpoint under consideration. * Age may be a risk factor for the behavioral effects of exposure to heavy particles and the effects are not necessarily linear. * There are changes in the expression of dendrites in the hippocampus as a function of exposure to HZE particles and these changes are related to changes in performance.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Articles in Peer-reviewed Journals	Rabin BM, Carrihill-Knoll K, Hinchman M, Shukitt-Hale B, Joseph JA, Foster BC. "Effects of heavy particle irradiation and diet on object recognition memory in rats." Advances in Space Research 2009 Apr 15;43(8):1193-9. http://dx.doi.org/10.1016/j.asr.2009.01.015 , Apr-2009
Books/Book Chapters	Rabin BM, Joseph JA, Shukitt-Hale B. "Effects of Berry Fruits on Neurocognitive Deficits Produced by Exposure to Space Radiation." in "Herbal Radiomodulators: Applications in Medicine, Homeland Defence and Space." Ed. R. Arora. Wallingford, United Kingdom : CABI Publishing, p. 151-162, 2008., Sep-2008